Optional Stopping for Bounded Stopping Times

An End-to-End Meta-Math Walkthrough

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Abstract

Stopping retains exactly those increments whose inclusion was decided before the next increment was observed. That predictability preserves their zero conditional means. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

PS-03 / Probability and Stochastic Processes / Report v1

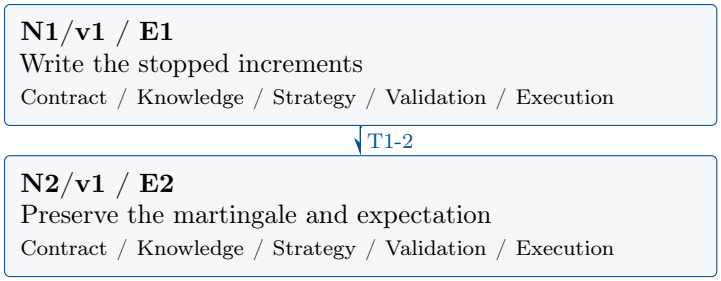
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $(M_n, \mathcal{F}_n)_{n\geq 0}$ be an integrable real martingale: M_n is $\mathcal{F}_n$-measurable, $\mathbb{E}|M_n| < \infty$, and $\mathbb{E}[M_n \mid \mathcal{F}_{n-1}] = M_{n-1}$. Let τ be a stopping time with integer values and $0 \leq \tau \leq N$ almost surely for some deterministic finite N. Prove $\mathbb{E}M_\tau = \mathbb{E}M_0$. Also prove $(M_{n\wedge\tau})$ is a martingale.*

A. Foundations and permitted imports

Use linearity and the tower property of conditional expectation, and the pull-out property for bounded measurable factors. A stopping time means $\{\tau \leq n\} \in \mathcal{F}_n$. Finite sums of integrable variables are integrable. The boundedness condition is on the time, not on sample values of the martingale.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $M_{n\wedge\tau} = M_0 + \sum_{k=1}^{n} D_k 1_{\{\tau \geq k\}}$, with each stopped increment conditionally mean zero.

Receiving activity. N2.1

Compatibility. E1/v1 and the deterministic bound $\tau \leq N$. Check: No limit of expectations at an unbounded time is taken; the final stopped variable equals $M_{N\wedge\tau}$.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Write the stopped increments

Object and role. Telescope the stopped path with indicators known at the preceding time. **Local obligation:** $M_{n\wedge\tau} = M_0 + \sum_{k=1}^{n} D_k 1_{\{\tau\geq k\}}$, with each stopped increment conditionally mean zero. **Inputs.** $D_k = M_k - M_{k-1}$ and stopping time $\tau \leq N$.

C — Contract

Publish E1/v1 only after establishing this obligation: $M_{n\wedge\tau} = M_0 + \sum_{k=1}^{n} D_k 1_{\{\tau\geq k\}}$, with each stopped increment conditionally mean zero.

K — Knowledge

$D_k = M_k - M_{k-1}$ and stopping time $\tau \leq N$. Retain the stated assumptions and import versions.

S — Strategy

Telescope the stopped path with indicators known at the preceding time.

V — Validation

$\{\tau \geq k\} = \{\tau \leq k-1\}^c$ belongs to $\mathcal{F}_{k-1}$; using $\{\tau = k\}$ instead would not have that predictability.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 On a path with $\tau = t$, the sum includes exactly increments 1 through $\min(n, t)$, so it telescopes to $M_{n\wedge t}$. The indicator $1_{\{\tau\geq k\}}$ is bounded and $\mathcal{F}_{k-1}$-measurable by the stopping-time property.

N1.2 The martingale property gives $\mathbb{E}[D_k \mid \mathcal{F}_{k-1}] = 0$. Pulling out the indicator yields $\mathbb{E}[D_k 1_{\{\tau\geq k\}} \mid \mathcal{F}_{k-1}] = 0$. Each product is integrable because its magnitude is at most $|D_k|$.

Export E1/v1. $M_{n\wedge\tau} = M_0 + \sum_{k=1}^{n} D_k 1_{\{\tau\geq k\}}$, with each stopped increment conditionally mean zero.

Acceptance gate. $\{\tau \geq k\} = \{\tau \leq k-1\}^c$ belongs to $\mathcal{F}_{k-1}$; using $\{\tau = k\}$ instead would not have that predictability.

Targeted repair. If a future-dependent indicator is pulled out of conditional expectation, replace it with the actual predictable inclusion event and verify its sigma-algebra. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Preserve the martingale and expectation

Object and role. Use the increment calculation for the stopped process, then evaluate at the finite horizon N. **Local obligation:** The stopped process is an integrable martingale and $\mathbb{E}M_\tau = \mathbb{E}M_0$.

Inputs. E1/v1 and the deterministic bound $\tau \le N$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: The stopped process is an integrable martingale and $\mathbb{E}M_\tau = \mathbb{E}M_0$.

K — Knowledge

E1/v1 and the deterministic bound $\tau \le N$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Use the increment calculation for the stopped process, then evaluate at the finite horizon N.

V — Validation

No limit of expectations at an unbounded time is taken; the final stopped variable equals $M_{N\wedge\tau}$.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 The finite-sum expression proves integrability and adaptation of $M_{n\wedge\tau}$. Its increment from $n-1$ to n is $D_n 1_{\{\tau\ge n\}}$, with conditional expectation zero by E1. Thus it is a martingale.

N2.2 Take expectations of the E1 identity at $n = N$. Each increment expectation is zero by the tower property. Since $N \wedge \tau = \tau$ almost surely, the left side is $\mathbb{E}M_\tau$, proving the equality.

Export E2/v1. The stopped process is an integrable martingale and $\mathbb{E}M_\tau = \mathbb{E}M_0$.

Acceptance gate. No limit of expectations at an unbounded time is taken; the final stopped variable equals $M_{N\wedge\tau}$.

Targeted repair. If boundedness of τ is removed, stop at $\tau \wedge n$ first and state the additional integrability condition needed for any subsequent limit. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Write the stopped increments

1. **N1.1.** On a path with $\tau = t$, the sum includes exactly increments 1 through $\min(n, t)$, so it telescopes to $M_{n\wedge t}$. The indicator $1_{\{\tau\geq k\}}$ is bounded and $\mathcal{F}_{k-1}$-measurable by the stopping-time property.

2. **N1.2.** The martingale property gives $\mathbb{E}[D_k \mid \mathcal{F}_{k-1}] = 0$. Pulling out the indicator yields $\mathbb{E}[D_k 1_{\{\tau\geq k\}} \mid \mathcal{F}_{k-1}] = 0$. Each product is integrable because its magnitude is at most $|D_k|$.

B. N2 — Preserve the martingale and expectation

1. **N2.1.** The finite-sum expression proves integrability and adaptation of $M_{n\wedge\tau}$. Its increment from $n-1$ to n is $D_n 1_{\{\tau\geq n\}}$, with conditional expectation zero by E1. Thus it is a martingale.

2. **N2.2.** Take expectations of the E1 identity at $n = N$. Each increment expectation is zero by the tower property. Since $N \wedge \tau = \tau$ almost surely, the left side is $\mathbb{E}M_\tau$, proving the equality.

□

V. WORKED ORCHESTRATION

Let M_n be the simple symmetric random walk started at zero, and let τ be the first time it reaches $+1$, capped at time two. The stopped values are $+1$ with probability $1/2$, 0 with probability $1/4$, and -2 with probability $1/4$. Their expectation is $1/2 - 1/2 = 0 = M_0$, matching the theorem.

A. Boundary and hypothesis audit

For fair coin tosses, set $M_n = 2^n 1_{\{\text{first } n \text{ tosses are tails}\}}$, so $M_0 = 1$. It is a nonnegative martingale because the next step doubles with probability $1/2$ and becomes zero otherwise. If τ is the first head, $\mathbb{P}(\tau > n) = 2^{-n}$, so $\tau < \infty$ almost surely, but $M_\tau = 0$ and its expectation is not one. Almost sure finiteness is insufficient.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 limit: $M_{\tau \wedge n} \to M_\tau$ almost surely, therefore their expectations converge. The doubling example has expectation one at every finite stopped horizon but limit expectation zero. **Diagnosis:** Validation finds an invalid exchange without uniform integrability or domination. **Repair:** use the stated deterministic bound N, avoiding the limit entirely. An unbounded-time theorem would require a separately proved stronger contract.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Extend to uniformly integrable martingales and unbounded stopping times, explicitly proving the expectation-limit passage.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Amir Dembo, Probability Theory: STAT310/MATH230 notes for measure-theoretic probability and limit arguments; Stanford STAT310C syllabus and lecture materials for martingale inequalities and stopping theorems. The versions and proofs used in this collection are stated explicitly.